

# Precautionary Electrification\*

Audrey Azerot<sup>†</sup>

October 18, 2025

[Latest version](#)

## Abstract

This paper introduces and quantifies a new mechanism, precautionary electrification, whereby households adopt electricity instead of gas to insure themselves against volatile gas prices. The analysis is motivated by two facts: natural gas prices in the United States have been far more volatile than electricity prices over the past two decades; and lower-income households are much more likely to rely on electricity for heating than higher-income households. Using state-level data from 1999 to 2023, I show that greater gas price volatility leads to higher electrification, particularly among low-income households. I develop a structural model of household energy choice with non-homothetic preferences and costly fuel switching. The model matches the empirical relationship between income, fuel choice, and energy expenditure, and implies that gas price volatility generates regressive welfare losses and a trade-off between short-term price insurance and long-term electrification incentives.

*Keywords:* electrification, energy price, households energy choice, risk.

*JEL codes:* D12, D81, E21, Q41, Q48.

---

\*I am indebted to my advisors Simon Gilchrist, Virgiliu Midrigan and Corina Boar for their generous guidance, encouragement and many insightful discussions. I also thank Tanvi Bansal, Jess Benhabib, Jaroslav Borovička, Katarina Borovičková, Nicolò Ceneri, Guzel Ishmaeva, Aya Jibet, Masayuiki Okada and Jean-Christophe Vergnaud for helpful comments and suggestions. I am grateful to seminar participants at the NYU Student Macro Lunch and the NYU Stern Lunch. Some of the data used in this analysis are derived from Restricted Data Files of the PSID, obtained under special contractual arrangements designed to protect the anonymity of respondents. These data are not available from the author. Persons interested in obtaining PSID Restricted Data Files should contact [psidhelp@umich.edu](mailto:psidhelp@umich.edu). All remaining errors are my own.

<sup>†</sup>New York University; email: [aa8010@nyu.edu](mailto:aa8010@nyu.edu).